

W.D. Chathunka J. Tennakoon

Koshigaya City, Saitama Prefecture, Japan.

+818067691992

cjtennakoon@gmail.com

[linkedin.com/in/chathunka-tennakoon.](https://www.linkedin.com/in/chathunka-tennakoon)

<https://github.com/Chathunka>



Working Experience



Senior Engineer (DX / xTECH Solutions Division / XTS: xTECH Solutions Department), BRYCEN Co., Ltd. Japan (May 2024 – Present)

- Designing and implementing advanced robotics systems, including SLAM, real-time control, and embedded GPU/FPGA-based computing (Jetson, Kria K26, ESP32, etc.).
- Developing firmware and utility software for embedded systems.
- Leading R&D initiatives focused on embedded robotics and AI-driven solutions.
- Developing web, mobile, and desktop applications to support embedded system applications and IoT solutions including control interfaces, data visualization tools, and diagnostic utilities.
- Overseeing team leadership and project management activities to ensure timely and efficient delivery of technical objectives.

 Consulting Engineer (IIoT), N-able Private Limited (Sept 2023 – March 2024)

- Head of the IIoT business unit
- Business development and Presales engineering.
- Team leading and project Management.
- IIoT Solution Design.
- Technology Evaluation.
- Business analysis.
- Training and Knowledge Transfer.



Internship (Quality Control in Embedded System Development), BRYCEN Co., Ltd. Japan (Aug 2023 – Oct 2023)

- Software Development (MEAN technology stack).
- Mobile Application Development (Flutter).
- Firmware & Utility Software Development (Embedded Systems).
- Team leading and project Management.

 Senior Engineer (IoT), N-able Private Limited (Mar 2023 – Sept 2023)

- Software Development (MERN technology stack).
- Firmware & Utility Software Development (Embedded Systems).
- Hardware development (Embedded System electronics).
- Team leading and project Management.
- Presales engagement.



Instructor, University of Colombo School of Computing (UCSC) (Feb 2022 – Mar 2023)

- Software Development.
- Firmware & Utility Software Development (Embedded Systems).
- Hardware development (Embedded System electronics).
- AI & ML program development.



Design & Operation Executive, United Motors Lanka (2020 – 2022)

- 3D Modeling, Printing, Visualization, and Architectural Rendering.
- Embedded Firmware & Software Development
- Project Management and Team lead.
- Presales engineering.



Lecturer in Aircraft Engineering & Maintenance, Sri Lanka Technical Training (STT), SriLankan Aviation College (SLAC) (2017 – 2019)

- Conducted lectures and practical sessions for Aircraft Maintenance Engineering Students (Computer Science, Software, Embedded systems).



Co-Founder & Software Developer, Internet of Things & Electronic Solutions (IOTES) (2012 – 2019)

- Software Development.
- Firmware & Utility Software Development (Embedded Systems).
- Hardware development (Embedded System electronics).

Education



Bachelor of Science (B.Sc. Special), Computational Physics, University of Colombo (2012 – 2017).



Diploma in Information Technology, University of Colombo School of Computing (UCSC).

Professional Qualifications and skills

- **Advanced program in Embedded Technology Engineer**, implemented by Association for Overseas Technical Cooperation and Sustainable Partnerships (AOTS) entrusted by the Ministry of Economy, Trade and Industry or METI, together with Japan Embedded Systems Technology Association (JASA). – **onsite training in JAPAN (Aug-2023 - Present)**
- Workshop on **Scientific Use of Machine Learning on Low-Power Devices**: Applications and Advanced Topics - Abdus Salam International Centre for Theoretical Physics (ICTP).
- **LFS110x: Business Considerations for 5G with Edge, IoT, and AI** (a course of study offered by LinuxFoundationX, an online learning initiative of The Linux Foundation).
- **GCCF1x: Google Cloud Computing Foundations: Cloud Computing Fundamentals** (a course of study offered by GoogleCloud, an online learning initiative of Google Cloud).
- **LFS151.x: Introduction to Cloud Infrastructure Technologies** (a course of study offered by LinuxFoundationX, an online learning initiative of The Linux Foundation).
- **Programming Languages**: C, C++, Python, Java, C#, PHP, JavaScript, TypeScript
- **Frameworks & Tools**: .NET Core, React, Flutter, Node.js, Express, OpenCV, MATLAB
- **Embedded Systems**: STM32, AVR, ESP32, Kria K26 (FPGA), Jetson Orin (GPU)
- **Communication Protocols**: CAN, Modbus, RS232, I²C, USB, One-Wire, Bluetooth, Wi-Fi
- **Embedded Firmware Development**: Extensive experience developing firmware in C, C++, and MicroPython
- **IoT & Robotics**: Skilled in Internet of Things (IIoT) solutions and robotic systems
- **Real-Time & Autonomous Systems**: Hands-on experience with ROS 2, micro-ROS, SLAM (2D/3D), and AI/ML model training pipelines

- **High-Performance & Intelligent Computing:** Background in parallel computing, data analytics, and neural computing
- **Computer Vision & Simulations:** Proficient in image processing, computer graphics, and simulations using OpenCV, OpenGL, and MATLAB
- **CAD & Prototyping:** Proficient with SolidWorks, Fusion 360, and AutoCAD for 3D design. Skilled in mechanical prototyping using lathe, milling, CNC tools, and 3D printers
- **Cloud & DevOps:** Experience with AWS, Google Cloud (Foundations), Git, Docker, and CI/CD pipelines

Projects and Activities

- **Autonomous Driving Perception – Road Segment Alignment, Anomaly Detection & 3D Annotation (Sony Honda Mobility Japan (SHM) – Autopilot Assistance Project July 2025 – Present)**

Developed and optimized a comprehensive perception data processing pipeline to support autonomous driving functions. Focused on aligning lane segments from multi-sensor data and preparing high-quality, annotated datasets for AI/ML model training and validation.

Key Highlights:

- Designed tools to read, process, and visualize JSON-based road segment data generated from 6 synchronized camera views and a 3D LiDAR system.
 - Built a high-precision lane segment alignment module to correct geometric inconsistencies and fuse multi-sensor data into accurate bird's-eye view (BEV) images.
 - Implemented algorithms for detecting missing or misaligned lane segments and identifying outliers to ensure annotation accuracy.
 - Developed GUI-based visualization and editing tools to support interactive segment correction and anomaly inspection.
 - Integrated 3D annotation support for labeling and visualizing spatial features critical for training autonomous driving models.
 - Curated high-quality training datasets by combining automated checks and manual refinement, ensuring suitability for ML/AI pipeline integration.
- **ROS 2 Robot Car with micro-ROS on ESP32-S3 – Japan (Mar 2025 – Present)**

Developing a real-time mobile robot control system by integrating ROS 2 Rolling on Raspberry Pi with micro-ROS running on an ESP32-S3-based motor controller.

Key Highlights:

- Implemented real-time robot motion control using geometry_msgs/Twist messages over UDP, enabling responsive and reliable wireless command execution.

- Designed and developed a lightweight command protocol over USB and UDP for flexible motor control, supporting direction and speed commands.
- Built and deployed micro-ROS firmware on ESP32-S3 for subscribing to ROS 2 velocity commands and translating them into motor driver actions.
- Integrated serial (USB) and network (UDP) communication modes to support both development and deployment scenarios.
- Tuned motor response and control loop timing to achieve smooth and accurate movement of the robot car in teleoperation and autonomous tasks.
- Path planning Frontier detection and Obstacle avoidance.

This project showcases practical experience in embedded ROS 2 communication, real-time robot control, network protocol design, and the integration of microcontroller-based systems into modern robotic architectures.

- **3D SLAM Porting for Kria K26 SOM (FPGA) and Jetson Orin Nano (GPU) – Japan (Nov 2024 – June 2025)**

Leading a high-performance embedded SLAM development project focused on porting and optimizing the 3D SLAM algorithm for both Kria K26 SOM (FPGA) and Jetson Orin Nano (GPU) platforms.

Key Highlights:

- Ported the 3D SLAM algorithm to run efficiently on Kria K26 SOM (FPGA) and Jetson Orin Nano (GPU), utilizing each platform's parallel processing architecture for real-time performance.
- Optimized memory usage and computational throughput to improve overall system responsiveness on resource-constrained hardware.
- Developed custom hardware accelerators for FPGA and CUDA-based GPU algorithms to enable fast and accurate 3D SLAM processing.
- Achieved real-time SLAM performance with reduced latency and improved mapping accuracy using LiDAR and RGB-D sensor fusion.
- Benchmarked performance, power consumption, and computational throughput across FPGA and GPU implementations to guide hardware selection for autonomous systems.

This project highlights expertise in FPGA and GPU system design, real-time algorithm deployment, and the integration of advanced SLAM techniques into embedded systems for robotics and autonomous navigation.

- **2DSLAM implementation and optimization (Google Cartographer) - May 2024 – June 2025 (Japan Onsite)**

Project focused on achieving precise 2DSLAM algorithm implementation and optimization.

Key highlights include:

- Reading and implementing Google cartographer method on 2D SLAM application.
- Optimizing precomputed submap creation method using alternative methods.
- Comparing and analysing Conventional Scan Matching (CSM) and Depth First Search Branch and Bound Scan-Matching (DFSBBBS) on real scanned data sets.

This project showcases expertise in advanced SLAM algorithms, spatial data analysis, and the ability to implement and optimize cutting-edge solutions for precise localization in dynamic environments.

- **Precision Indoor UWB Localization - Aug 2023 - Oct 2023**

Project focused on achieving precise indoor localization using Ultra-Wideband (UWB) technology.

Key highlights include:

- Utilizing trilateration techniques for pinpoint location accuracy in indoor environments.
- Integration of ESP32 devices for seamless connectivity and user-friendly configuration.
- Development of a mobile app for real-time data visualization and a web app for expanded accessibility.

This project showcases expertise in cutting-edge technology and the ability to develop user-friendly solutions for embedded systems, mobile and web interfaces.

- **Dangerous Gas Monitoring Solution (Mar 2023 – Mar 2024 [Industry])**

This comprehensive system involved strategically installing gas sensors for dangerous gases like (CO, H₂S, Cl₂, CH₄, etc.), establishing a robust Modbus communication network, and creating a custom IIoT Gateway for real-time data collection and transmission at one of the main container ports in Sri Lanka. Leveraging the MERN technology stack, a cloud-based platform (AWS) was designed for data visualization, analysis, and management, enabling users to monitor gas levels, set alarms, and receive immediate notifications. The solution significantly enhanced the safety, operational efficiency, and data-driven decision-making of port operations.

- **Smart office (automation Mar 2023 – Mar 2024 [Industry])**

Designed and implemented a Smart Office IoT solution with real-time O₂, CO₂ monitoring, alarm generation, and ventilation control. Integrated IP camera-based real-time human detection for efficient lighting and AC management. Developed a user-friendly mobile app for equipment control, optimizing employee comfort and energy savings.

- **IoT smart garden controller and switch (Feb 2022 – Mar 2023 [Industry])**

This is an ongoing project called “IoT Smart Garden Controller and Switch” which combines cutting-edge technologies, including ESP32, Flutter, and .NET Core, to provide a comprehensive solution for managing a garden's irrigation system. The project can help individuals, communities, and businesses to conserve water, reduce waste, and improve the health and productivity of their gardens.

- **IoT Greenhouse automation system (Feb 2022 – Mar 2023 [Industry])**

This is an ongoing project in which I am involved as a researcher in the UCSC research group and the department of Plant Science at the University of Colombo. The project is focused on developing an IoT-based Greenhouse Automation System. The system is based on the Arduino ESP 32 platform, and is able to record sensor data on an SD card and send it to Google Sheets. The project employs various sensors to measure parameters such as NPK, EC, light intensity, humidity, and temperature.

- **Vehicle tracking system (Fleet Tracking with GPS and other sensor data [Industry])**

A fleet tracking system with location, fuel level, engine status, and other sensory details with geofencing is a comprehensive solution for monitoring and managing a fleet of vehicles. The system uses GPS technology and a variety of sensors to gather data on the vehicles' location, speed, route, fuel consumption, engine status, and other important metrics.

The system also includes geofencing capabilities, which allow administrators to define virtual boundaries around specific locations. When a vehicle enters or leaves a geofenced area, the system generates an alert, providing real-time information on the vehicle's movements.

In addition to geofencing, the system may also provide real-time alerts for various events, such as low fuel levels, speeding, or engine issues. This information can be accessed through a web-based platform or a mobile app, and the system can be integrated with other software tools, such as dispatch and routing software, to enhance fleet management and efficiency.

<https://www.youtube.com/watch?v=pjLLVso4FRQ>

- **GPRS/GSM Cooler Data Logger & Remote Terminal Unit [Industry]**

GPRS/GSM Cooler Data Logger & Remote Terminal Unit is a device which can be used to acquire data such as Temperature, Vibrations, Noise level, Power, Voltage, and Current of the compressor, etc. related to any cooler device.

- **Wi-Fi / 3G Telemetry Data Logger & Remote Terminal Unit for bear flow control system. [Industry]**

Wi-Fi / 3G Telemetry Data Logger & Remote Terminal Unit is a very user-friendly device which can be used for acquiring data from the particular environment system using standard sensor units in order to upload data to backend server and then analyze them to generate reports.

- **Image Processing and Object detection (Hand drawn ER diagram to technical relationships with SQL statements [Research])**

Program is to convert hand drawn Entity Relationship Diagrams (hardcopies, camera images) to digital format (softcopies) using open CV with python.

- **Fully automatic Tennis Ball Shooter [Industry]**

The "Tennis Ball Shooter" offers a comprehensive training solution for professional tennis players, combining speed control, angle adjustment, and ball spinning options with an automatic ball feeder and remote control.

- **Image processing / Neural Computing / Computer Graphics [Research]**

- Currency detection.
- Graphical simulation of a 3D model house using Open GL and C#.

- **Other projects**

- Building Management System (Home automation)
- Fingerprint scanning and RFID system with the database management using C#
- Student data and attendance management system.
- WIFI display unit.
- IOT Smart switches and electric sockets.
- Vending machine controller.
- Item Maintenance Management System

Extra-Curricular Activities

- Qualified Member of Life Saving Association Sri Lanka.
 - Basic rescue
 - Elementary Swimmer GR:3
 - Resuscitation
 - Intermediate
- 2015-2016 First President of the “Colombo Innovations and Robotics Club (CIRC)” University of Colombo Faculty of Science.
- 2015-2016 Editor (4th year), Physics Society (PhySoc) of University of Colombo Department of Physics.
- 2014-2015 Main organizer (3rd year), Physics Society (PhySoc) of University of Colombo Department of Physics.
- 2014-2015 Network Administrator of the Students' Council Computer Lounge, University of Colombo, Faculty of Science.
- Preveshika Examination in Vocal with second division from Bhatkhande Sangit Vidyapith Lucknow.
- Sports: Swimming, Chess with awards.

References

Updated references, including those from recent and previous roles, are available upon request.